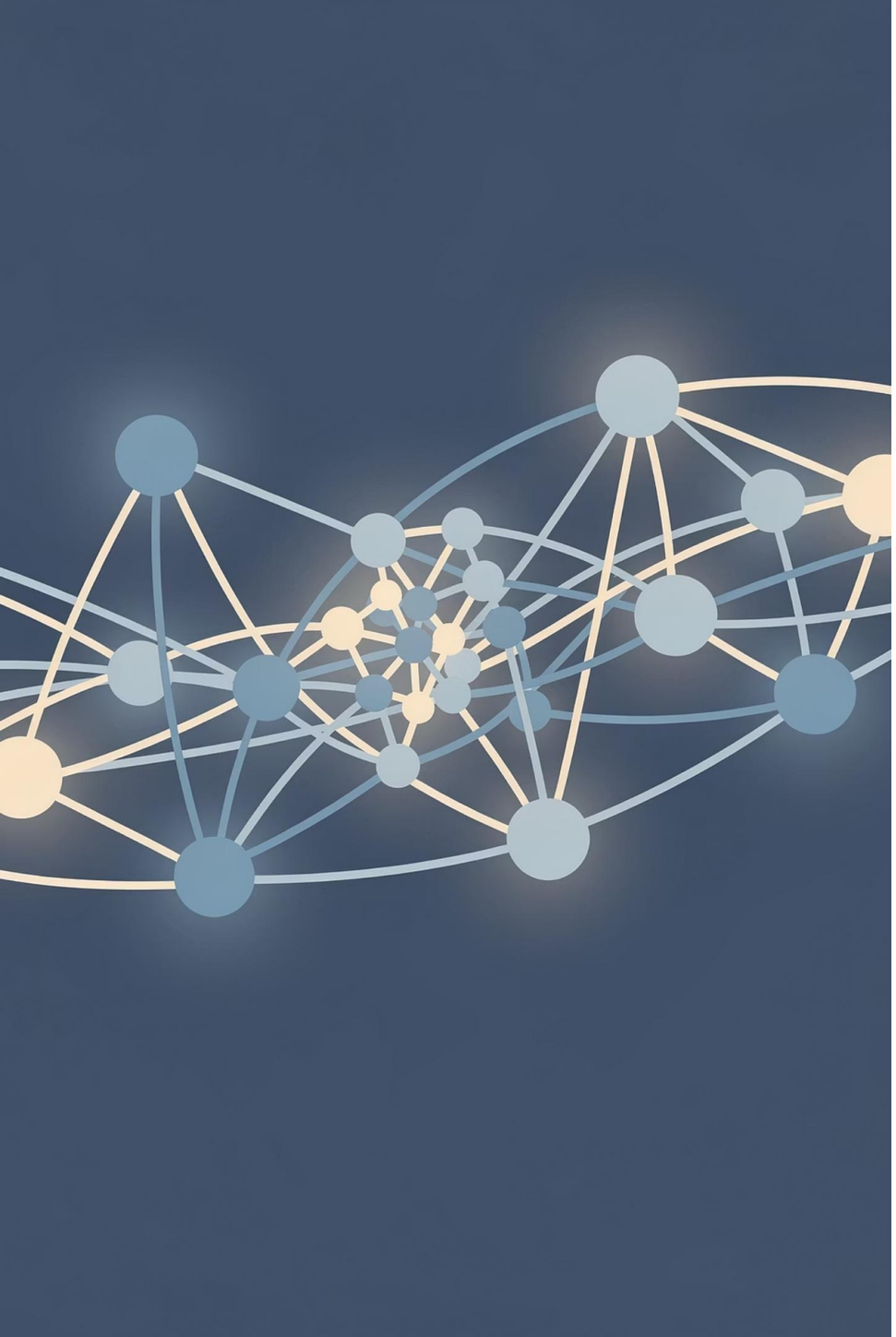
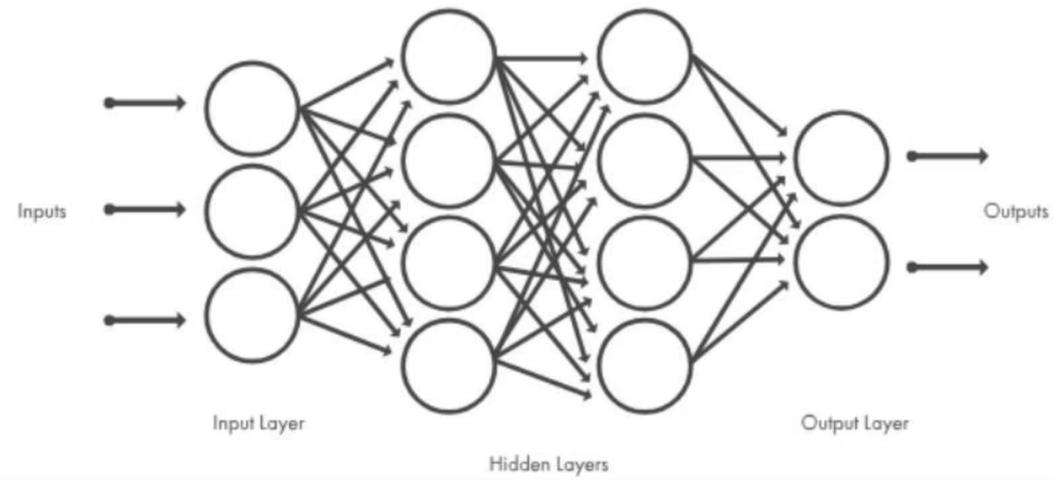


Neural Network Basics
Image classification example

A summary from Deep Leaning specialization Course by Andrew NJ

ARCHITECTURE

Neural Network Structure



Neural networks consist of interconnected layers: an Input Layer receives data, Hidden Layers process information through weighted connections, and an Output Layer produces predictions. Every node connects to every node in the next layer, creating a dense network of computations.

Summarized by : Joracin

Matrices is Everything!

$$\begin{pmatrix} -8 & 5 & 1 & 7 & 2 \\ 0 & 1 & 3 & 5 & -6 \\ 2 & 2 & 1 & 5 & 3 \\ 0 & 4 & 3 & 7 & 2 \\ 1 & 1 & -7 & -6 & 5 \end{pmatrix}$$

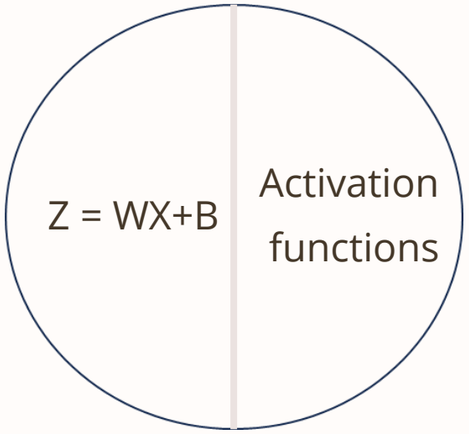


Image Input

Photos are converted into numerical matrices where each pixel becomes a value

Matrix Representation

Pixel intensities form the foundation for neural network processing

Neuron Processing

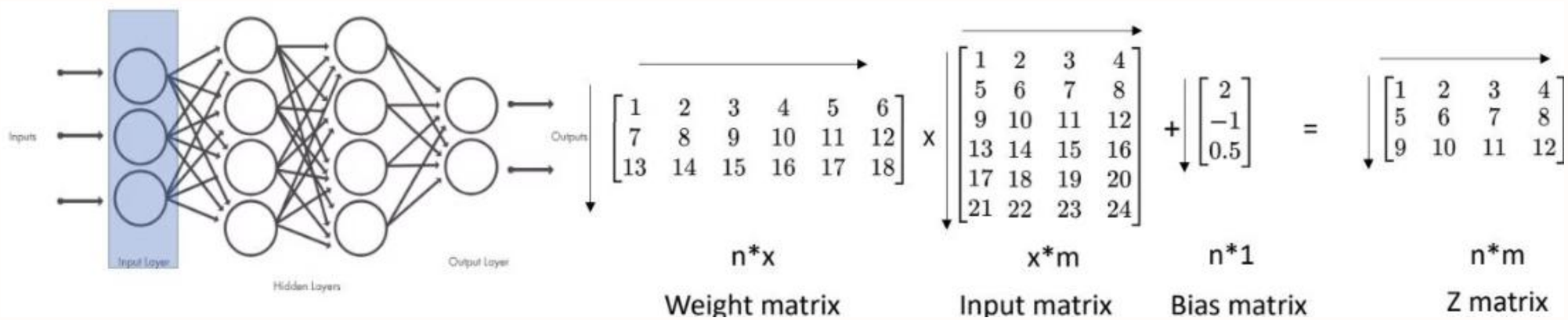
Each neuron applies Z = WX+B then activation functions

Summarized by : Joracin

Matrix Mathematics in Action

Matrix Structure

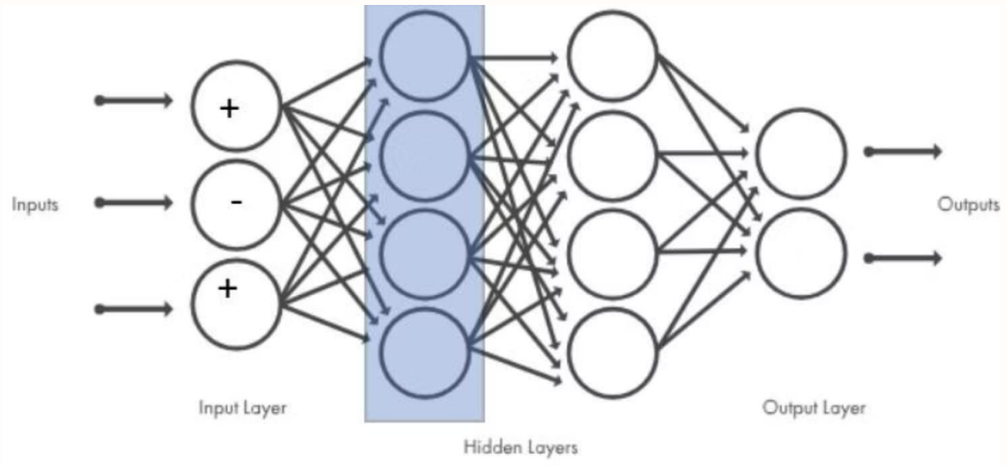
For 3 neurons with 6 inputs and 4 training examples:



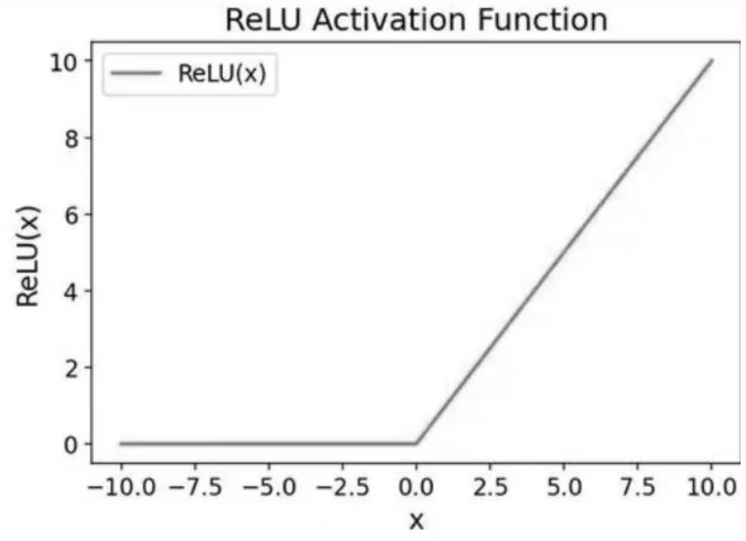
Dimensions: Weight (n×6) × Input (6×m) + Bias (n×1) = Output (n×m)

Summarized by : Joracin

Feature Weighting and Activation



Each neuron has different weights assigned to same set of features, highlighted nodes show active pathways



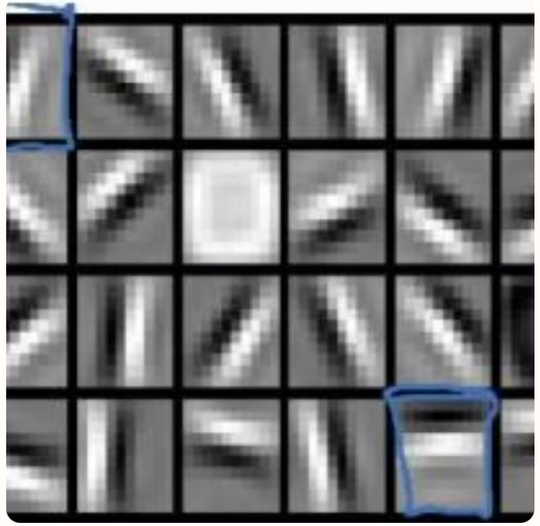
ReLU Activation Function: Zero for negative inputs, linear for positive values. Determines which neurons activate and pass signals forward.

Summarized by : Joracin

DEEP LEARNING

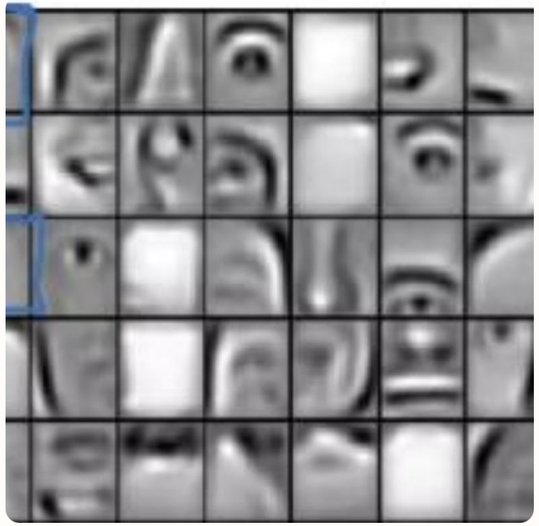
Progressive Feature Learning

Neural networks learn increasingly complex features across layers, from simple edges to complete objects.



Layer 1: Basic Patterns

Detects edges, lines, and simple shapes



Layer 2: Facial Features

Recognizes eyes, noses, and facial components

Layer 3: Complete Faces

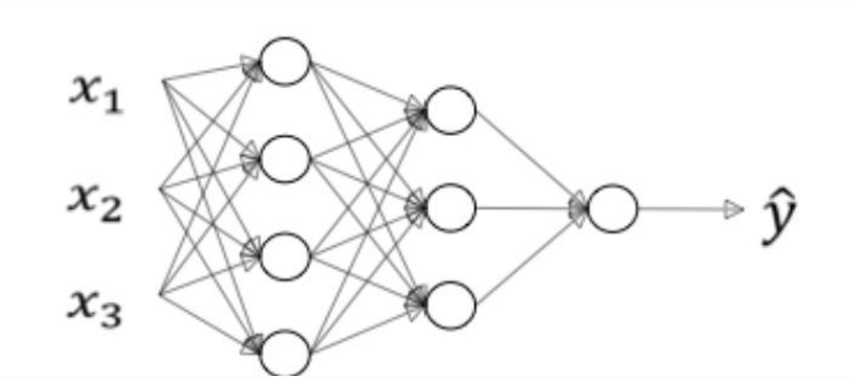
Assembles features into recognizable faces

Summarized by : Joracin

Training: Learning from Examples

Truth label = 1

Truth Label = 0



Initial Prediction **0.67**

Actual Truth **0**

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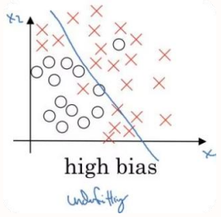
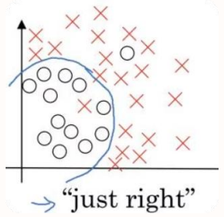
Bias vs. Variance Problem



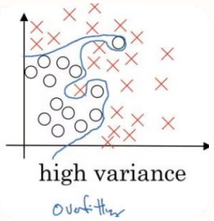
High Bias (Underfitting)

Problem: Model too simple to capture patterns



Just Right

Model generalizes well to new data with appropriate complexity and training



High Variance (Overfitting)

Problem: Model memorizes training data

Summarized by : Joracin

Optimization Techniques

Addressing Bias

Bigger network, train longer, different architecture

Data Augmentation

Crop, skew , tilt to create more data for training

Regularization

Penalizes large weights to prevent overfitting

Dropout

Randomly skips neurons during training to improve generalization

Summarized by : Joracin

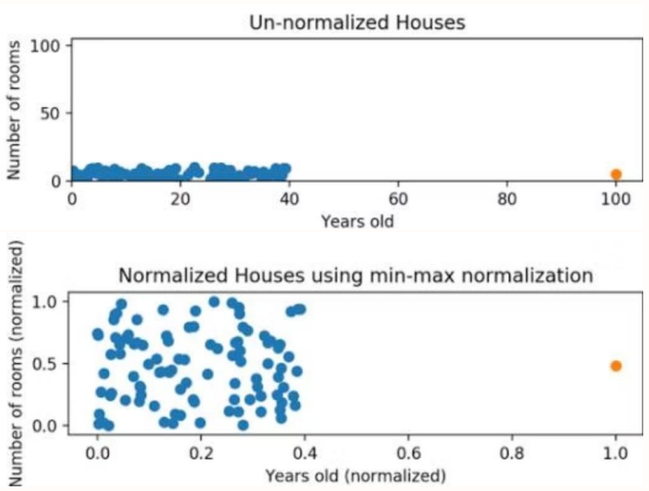
Optimization Techniques

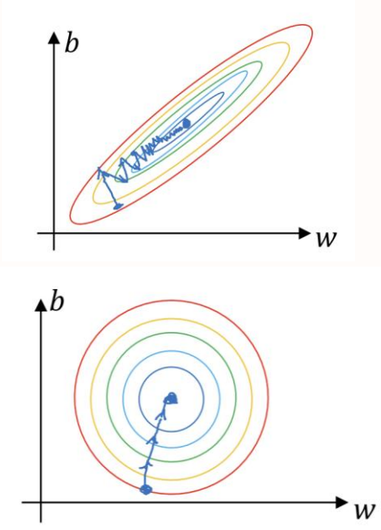
Normalization

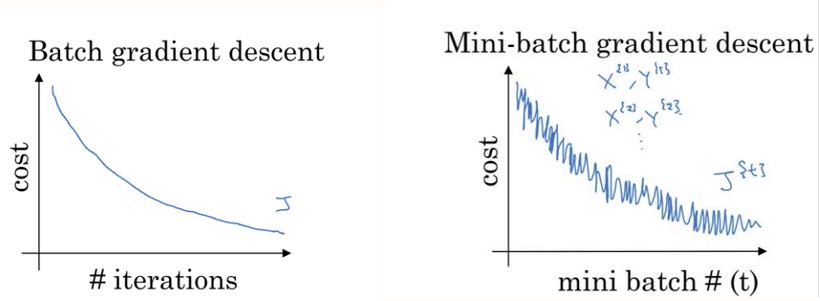
Scales features to similar ranges, accelerating training convergence

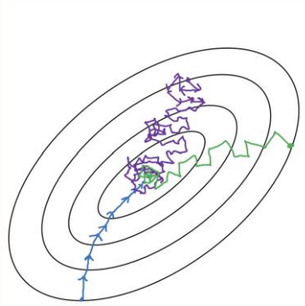
Mini-Batch Gradient Descent

Processes data in small batches for faster, more stable learning









Summarized by : Joracin

Hyperparameter Tuning Strategy

1

Highest Priority

- Learning rate
- Number of hidden units
- Batch size
- Momentum/Adam beta values

2

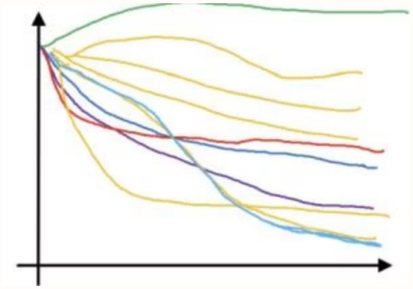
Lower Priority

- Regularization
- Number of layers
- Activation functions
- Mini-batch size

Two Training Approaches

Babysitting One Model: Carefully monitor and adjust a single model's training, iterating on hyperparameters sequentially.

Training Many Models in Parallel: Run multiple configurations simultaneously to find optimal settings faster.



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